

Creating Creature Chaos: The Methods that Brought Crowds to the Forefront on Disney's "Strange World"

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Figure 1: A shot showcasing Goblinswills, Terrordactyls, and Reapers



Figure 2: A shot from the "Chase in the Windy Jungle" sequence ©Disney

ABSTRACT

The fantastical, frenzied environment of Walt Disney Animation Studios' 61st film "Strange World" necessitated an ecosystem teeming with unique creatures that form the literal lifeblood of the film. Multiple new tools and pipeline optimizations were required to handle the large scope of the project. We present a hybrid Crowds system combining Pixar's UsdSkel [Pixar 2023] and a USD point-based caching system that uses our in-house instancer Aurora. Our method combines the flexibility of skeletal manipulation of character animation cycles for foreground elements with robust scalability to allow for millions of characters in the background. In addition, we describe tool sets for the creation of notable creature performances including procedural "Goblinswill" streams, a flocking system for "Terrordactyls," and a swarming system for "Reapers."

CCS CONCEPTS

• **Computing methodologies** → **Procedural animation**; *Physical simulation*.

KEYWORDS

Crowds, LOD, Pipeline, Procedural Geometry, Simulation

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1 INTRODUCTION

Crowds creatures exist in enormous numbers throughout Disney's "Strange World." These creatures play a core role in the story, at times driving the hero animation and cinematography. Enabling this required building on existing Crowds tools and processes. The Crowds Department has been utilizing Pixar's UsdSkel since the studio pipeline adopted USD on "Encanto." To support both the needed control for foreground elements and the scalability required for background elements, we developed a hybrid system leveraging both the flexibility of UsdSkel skeletal manipulation and the efficiency of point-based caching using our Aurora tool set. Additionally, many tools were developed to produce various creature-specific behaviors by combining the power of our in-house Crowds system Skeleton Library [El-Ali et al. 2016] and Houdini.

2 PIPELINE CHANGES

2.1 Departmental Collaboration

Historically, the Crowds department begins their work after Layout and Animation have completed. However, significant modification to this workflow was required for the key sequence "Chase in the Windy Jungle." In this sequence, our protagonists are pursued along treadmill-like streams formed by thousands of blood cell

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creatures known as "Goblinswills," as seen in Figure 2. To compose a shot, Layout constrained temporary Crowds assets published beforehand on art-directable splines. Then, to allow for proper foot contact, Crowds published a small section of blood cell streams and tuned their velocities for the Animation department. Next, in order for the animators to work with many thousands of blood cells, Crowds provided a lightweight mesh representation to act as a "ground plane." A secondary blood cells pass then assimilated mid or background blood streams into the shot.

2.2 Control and Scalability

UsdSkel supports flexible rig manipulation for character visualization and rendering as well as coherent data transfer between departments, but presented us with significant limitations to scalability. In our workflow, Crowds animation is authored in Houdini using animated clips or "cycles" via our in-house procedural skeletal system "Skeleton Library." Our prior implementation of UsdSkel supported only roughly a few hundred characters per shot, depending on their complexity. Due to the large number of creatures in "Strange World," a series of optimizations to the USD pipeline were required. Firstly, to reduce the amount of stage recomposition, frame independent data is written once and frame dependent data is queued and batched in a multi-threaded manner, then written in a single-threaded manner only when needed. All authoring is done using the Sdf API to avoid unnecessary intermediate recomposition. These changes made our UsdSkel transformations scale approximately linearly, giving us a nearly 40x speed-up in some cases. This increase in efficiency allowed for skeletal manipulation of thousands of characters which was sufficient for close up shots. When less control was necessary, Aurora was employed. This permitted greater scalability at a cost of reduced flexibility and skeletal rig control. Using this method, millions of creatures were instanced using pre-baked animation caches.

3 TOOLS AND CASE STUDIES

In total, the Crowds team was responsible for delivering the animated performances of 13 different types of creatures as well as humans and flying vehicles. The following sections describe three key creature behaviors in greater detail. (See Figures 1 and 3.)

3.1 Goblinswill Blood Streams

The Goblinswill blood cell streams feature in over 100 shots throughout the film, frequently in a prominent role or interacting with the film's protagonists. Due to the creative importance of these streams, it was necessary to develop an approach that offered precise art-directed control while still producing a consistent look in an efficient manner. To this end a procedural system using frame-agnostic, kinematic principles was built to allow fast, predictable and customizable calculation for each blood cell forming the stream. To



Figure 3: Goblinswill, Terrordactyl, and Reaper creatures ©Disney

prevent collisions between the highly dense blood cells as they flow, a procedural module provides separation by interactively squashing and stretching each blood cell in response to nearby neighbors. An ancillary set of tools then facilitates refining individual blood cell performance and composition. All shots featuring the blood cell streams made use of this rig. In most cases the procedural solution was sufficient. For rare cases requiring more complex behaviors, the procedural rig was still valuable as a first pass and the results then served as input to simulation-based techniques.

3.2 Terrordactyls

The Terrordactyls were inspired by biological killer T cells. Those controlled by Crowds needed to be able to emulate the motion complexity of the Hero-animated creatures while also handling large numbers and organic variation of movement. In order to achieve fluid, flight-like motion for these characters at scale, the following procedural tools were developed. For individuals, a spline-based flow, pitch, and roll control system was created. For large flocks, a creative repurposing of a Vector Field-based system was used. Finally, to add an additional level of realism to the vast flocks, a procedural bias scalar algorithm for calculating vector-determined tilt and roll was implemented.

3.3 Reapers

The deadly Reapers were based on macrophages, the hunters of the immune system, and needed to move with fine directional control but also with fluid, coordinated motion. To achieve this, a custom multi-solver simulation tool was implemented which achieves an insect-like swarming effect. A base layer of the Reapers is first simulated on the environment proxy geometry. This initial simulation pass provides gross frame-by-frame speed, direction, and position. A 3D locator and 2D flow field approach are combined in this step to enable control over directional movement while minimizing collisions. Post-Sim, a custom RBD-based stacking algorithm [Nghiem 2021] is used to further refine the look. Houdini's Vellum is then used in a final pass to clean up any remaining collisions.

4 CONCLUSIONS AND FUTURE WORK

There were over 56 million individual crowds elements featured in 28 percent of the total shots of "Strange World." This prominence of crowds creatures necessitated significant expansion of our pipeline and processes. The result was the combination of high fidelity with massive scale allowing for wondrous streams, flocks, and swarms of fantastic creatures to make this world truly strange. To build on this work, we next look to investigate the possibility of a new data representation that would seamlessly combine the power of UsdSkel with the scalability of Aurora instancing.

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